Wanqi Huang

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Education

National University of Singapore Aug. 2023–Present
Bachelor of Engineering in Electrical Engineering (Honours)

- Cumulative GPA: 4.93 / 5.0

University of California, Berkeley - Summer Session Jul. 2024

- Cumulative GPA 4.0/4.0

University of Cambridge - Engineering Exchange Program Oct. 2025–Jun.2026

- Selected as one of the three students for the one-year engineering exchange program at the University of Cambridge and completed ten advanced engineering modules
- Achieved First Class results

Achievements and Awards

- 2024/2025 Semester 1 and Semester 2 Dean's List
- Second Place in Cambridge AI Hackathon
- Second Place in IEEE-HKN Hackathon

Publications

- **Huang, W.**, Chen, X., Du, J., & Chua, D. (2026). Cascaded machine learning algorithm for snoring detection based on audio signal. In Y. Luo (Ed.), Cooperative design, visualization, and engineering (Lecture Notes in Computer Science, Vol. 16092, pp. 77–85). Springer. https://doi.org/10.1007/978-3-032-06952-8_9

Project Experience

Bio-inspired Clustering and Feature Discovery Algorithm Apr. 2026- Jun. 2026

*Researched on **bio-plausible feature discovery algorithm** in Cambridge Computational and Biological Learning Lab with professor Máté Lengyel*

- Developed clustering algorithm utilizing Hebbian/Anti-hebbian learning rule with sparsity constraints, enabling automatic sparse feature discovery
- Utilized Just-in-time (JIT) compilation to increase the speed of the algorithm by 5 times.

Ada-GO: Adaptive Global Optimisation for Co-Evolving Hardware and Policy Learning Dec. 2025- Apr. 2026

*Conducted Robotics hardware and **simulation** research at Cambridge Resilient Autonomous Learning Lab with professor Rika Antonova*

- Developed and improved simulations of hybrid-material robotic systems in different robot simulation software such as MuJoCo and Genesis.
- Developed an automatic pipeline for non-convex mesh decomposition, achieving accurate collision detection in MuJoCo simulation environment

Machine Learning-Based Snoring Detection Dec. 2024-Oct. 2025

***First-author** paper: "**Cascaded Machine Learning for Snoring Detection Based on Audio Signal**," presented at CDVE 2025. Supervised by Professor Chua Dingjuan*

- Extracted acoustic features from snoring signals using multiple signal processing techniques for tailored model training and analysis
- Designed a cascaded machine learning structure for snoring detection, achieving 95%accuracy
- Deployed the system on hardware for real-time snoring detection

Embodied AI with human fingertip inspired hybird robotic gripper for Surgical Inscrument Manipulation and Inspection Jan. 2024-Dec. 2024

*Built an **intelligent robotic system** with **hybrid-material fingertips** that can grip, manipulate, and inspect surgical instruments and classify them according to their type and cleanliness*

- Programmed an industrial robotic arm (xArm) to pick, manipulate, and place different types and randomly placed surgical instruments with the aid of the computer vision system, and automated the surgical instrument inspection and classification process
- Created a dataset with around 400 images of clean and contaminated surgical instruments of different types and trained a computer vision Yolo v8 model that achieved an **accuracy of 98%** under various lighting conditions
- Researched in soft robotics and 3D printed with soft TPU material and using silicone casting to create a hybrid gripper that can conform to various shapes of surgical instruments and pick them up from flat surfaces, outperforming commercial grippers

Lightweight Robotic Car for Obstacle Traversal Sep. 2023-Dec. 2023

*Crafted a robotic car that can complete the specified missions. **Broke the record** of the lightest weight robotic car in the **past three batches of students***

- Used Fusion 360 software to laser-cut wheels that can go through slopes and bumps
- Utilised ultrasonic sensors and Arduino board programming to control the robotic car's movement with marginal error within three centimetres
- Iterated multiple rounds to reduce the robotic car's weight, breaking the record of the lightest weight with a full functionality car

Professional Experience

NuSpace Pte. Ltd., AI Engineer Intern May. 2025-Aug. 2025

- Created a dataset and developed a computer vision model for satellite signal detection, achieving **95% accuracy.**
- Automated the satellite Doppler shift extraction process by replacing the manual signal labelling process and integrating the computer vision model, **doubling the efficiency** of satellite Doppler shift data collection
- Researched satellite orbit determination methods based on Doppler shift and prototyped approaches using grid search, particle filter, and Nelder–Mead optimisation, **reducing location error by 70% and velocity error by 65%**

Technical Skills

- **Programming Languages**: Python, Verilog, MATLAB, R
- **AI & Robotics:** Robotics Simulation, Machine Learning, Ubuntu, ROS2, Raspberry Pi, Arduino
- **Hardware & Prototyping:** CAD modelling, Silicone casting, 3D printing, Circuit Design

Co-curricular Activities

President of the Cooking and Baking Arts Society Oct.2025-Jun.2026

- Organised and managed society activities, coordinating logistics, ingredients, and participants

Vice-Captain for King Edward VII Soccer Team Aug.2024-May 2025

- Organised training twice a week and led a team with around 20 members
- Achieved first place in the 2024 NUS inter-hall competition